3.1.2 SPECIAL TOOLS

Some airbag diagnostic tests use special tools, the 8310 and 8443 airbag load tools for testing squib circuits. The load tools contain fixed resistive loads, jumpers, and adapters. The fixed loads are connected to cables and are mounted in a storage case.

The cables can be directly connected to some airbag system connectors. Jumpers are used to convert the load tool cable connectors to other airbag system connectors. The adapters are connected to the module harness connector to open shorting clips and to protect the connector terminal during testing.

When using the load tool, follow all of the safety procedures in the service information for disconnecting airbag system components. Inspect the wiring, connector, and terminals for damage or misalignment.

Substitute the airbag load tool in place of a Driver or Passenger airbag, curtain airbag, clockspring, or seat belt tensioner, (use a jumper if needed). Then follow all of the safety procedures in the service information for connecting airbag system components.

Read the module active DTC's. If the module reports NO ACTIVE DTC'S, the defective component has been removed from the system and should be replaced. If the DTC is still active, continue this process until all of the components in the circuit have been tested.

Then disconnect the module connector and connect the matching adapter to the module connector. With all airbags disconnected and adapter installed, the squib circuits can be tested for open and shorted conditions.

3.1.3 DIAGNOSTIC TROUBLE CODES

Airbag diagnostic trouble codes (DTC) consist of active and stored codes. If more than one DTC exists, diagnostic priority should be given to the active code(s). Each DTC is diagnosed by following a specific testing procedure. The diagnostic test procedures contain step-by-step instructions for determining the cause of the DTC. It is not necessary to perform all of the tests in this manual to diagnose an individual DTC.

Always begin by reading the DTC's using the DRBIII®. If more than one code exists, diagnostic priority should be given to active code(s).

Active DTC's for the airbag system are not permanent and will change the moment the cause of the code is corrected. In certain test procedures within this manual, DTC's are used as a diagnostic tool.

3.1.3.1 ACTIVE CODES

An active trouble code indicates an on-going malfunction. This indicates that the defect is currently active every time the ACM checks the particular circuit or function. It is impossible to erase an active code; active codes automatically erase themselves when the cause for the code has been corrected.

With the exception of the warning indicator trouble codes or malfunctions, when a malfunction has been detected by the ACM, the Airbag warning indicator is illuminated for a minimum of 12 seconds, or as long as the malfunction is present.

3.1.3.2 STORED CODES

Airbag codes are automatically stored in the ACM memory as soon as the malfunction is detected, with the exception of the Loss Of Ignition Run-Only code, which is an active code only. A stored code indicates that there was an active code present at some time. However, the code currently may not be present as an active code, although another active code may be.

When a trouble code occurs, the Airbag warning indicator illuminates for 12 seconds minimum, (even if the condition existed for less than 12 seconds). Stored codes display the time in minutes that the code was active, and the number of times that the ignition has been cycled since the active code was last detected.

The minimum time shown for any code will be one minute, even if the code was actually present for less than one minute. Thus, the time shown for a code that was present for two minutes 13 seconds, for example, would be three minutes.

If a malfunction is not active while performing a diagnostic test, the active code diagnostic test will not locate the source or the condition. In this case, the stored code can indicate an area to inspect. The following procedure may uncover a malfunction that is difficult to locate:

WARNING: TO AVOID PERSONAL INJURY OR DEATH, MAINTAIN A SAFE DISTANCE FROM ALL AIRBAGS WHILE PERFORMING THE FOLLOWING INSPECTION:

If no obvious problems are found:

- Erase the stored codes
- Place the ignition in the Run position
- Wiggle the wire harness and connectors
- Rotate the steering wheel from stop to stop
- Recheck for active codes periodically as you work through the system.